

WASIN WANGSOMBUT

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SUMMARY

Experienced Data Engineer with 8+ years of comprehensive background in designing and implementing data platforms across telecommunications, banking, and retail industries. Proficient in cloud-native data technologies including real-time streaming pipelines, lakehouse architecture, and Change Data Capture (CDC) patterns on Google Cloud Platform and Azure. Strong focus on scalable architecture, data quality, and cross-team collaboration to deliver production-grade data-driven solutions.

SKILLS

- **Programming Languages:** Python, Node.js (React), Pyspark
- **Web Development:** React, Next.js
- **Databases:** SQL Server, PostgreSQL, Hive, Teradata
- **Data Engineering:** ETL Development, Data Pipeline Design and Implementation (Streaming & Batch), Data Modeling, Data Validation, Schema Synchronization, Data Profiling
- **Stream Processing:** Apache Beam, Google Cloud Dataflow, Apache Kafka (Confluent Cloud, Schema Registry, Avro), Google Cloud Pub/Sub
- **Lakehouse & Storage:** Apache Iceberg, BigLake Metastore (BLMS REST Catalog), Google BigQuery, Google Cloud Storage, AWS S3 (Parquet)
- **Big Data Technologies:** Azure Databricks, Databricks Workspace API, Data Factory API, HD Insight, Unity Catalog
- **Cloud Services (GCP):** Dataflow (Flex Templates), BigQuery (Storage Write API, CDC), Cloud Bigtable, Secret Manager, Artifact Registry, Cloud Scheduler, Cloud Composer (Airflow)
- **Cloud Services(Azure):** Azure Data Factory, Azure Synapse Analytics, Azure Storage Account, Azure Key Vault
- **Infrastructure & DevOps:** Terraform (IaC), GitLab CI/CD, Docker, Kaniko, Jenkins, GitHub Workflow
- **Data Processing and Analysis:** PySpark, PySpark SQL, Kafka, Confluence
- **Automation and Scripting:** Automated Testing, Script Generation
- **Data Security and Governance:** Role-Based Access Control (RBAC)
- **CI/CD:** Jenkins , Github Workflow
- **Version Control:** Git (GitHub, GitLab, GitHub Enterprise Cloud)
- **Web Crawling:** Web Crawler Development
- **Testing:** Unit Testing, Integration Testing, Test Scenario Definition
- **Interpersonal Skills:** Collaboration with Data Scientists and Marketing Teams, Technical Support and Troubleshooting, Ad-hoc Issue Investigation and Resolution

WORK EXPERIENCE

Data Engineer , **June 2017 - December 2018**
Advanced Info Services Public Company Limited (AIS)

- Explored data from various sources to support the Data Science team in training recommendation models and creating data chains.
- Gathered data using web crawlers to support new marketing campaigns.
- Implemented ETL processes with SQL/Hive.
- Participated in data requirements discussions with Data Scientists and the Marketing team.
- Implemented data flow, data pipeline, and data cleansing using Databricks.

Data Engineer , **January 2019 - March 2019**
Total Access Communication Public Company Limited (DTAC)

- Developed ETL jobs using Teradata, ensuring efficient data integration and processing capabilities.

Frontend Developer, **April 2019 - December 2019**
Zimple Media Company Limited

- Developed web applications and blogs using React, Next.js, PostgreSQL, and GraphQL.
- Defined and performed test scenarios to ensure robust and user-friendly applications.

Data Engineer, **January 2020 - December 2022**
The Siam Commercial Bank Public Company Limited

- Designed and implemented ETL and data pipelines.
- Implemented data ingestion processes.
- Performed unit testing and integration testing.
- Developed job scheduling routines to ensure data quality and availability for analytical purposes.

Data Engineer, **January 2023 - April 2025**
SCB Data-X

- Designed and developed data models using PySpark and PySpark SQL on Azure Databricks.
- Concluded data validation processes.
- Developed automated tests and generated scripts using PySpark, Databricks Workspace API, and Data Factory API.
- Designed and developed schema synchronization processes using PySpark on Databricks to ensure data consistency across databases and data warehouses.
- Conducted POCs for the Databricks data platform migration to new versions using Unity Catalog.
- Developed RBAC processes to enhance data security.
- Profiled data sources in Kafka and Confluence for integration into new platforms.
- Investigated and supported ad-hoc issues, collaborating with platform teams to troubleshoot and resolve issues promptly.

WORK EXPERIENCE

Data Engineer (Consultant),
NTT DATA — Client: The 1 Central Group

May 2025 - Present

- Real-time streaming pipeline architecture: Confluent Kafka → Apache Beam (Dataflow) → two-layer lakehouse (Apache Iceberg as source of truth, BigQuery as refined/analytics layer).
- Batch pipeline architecture: daily incremental loads from PostgreSQL and REST APIs into the same Iceberg + BigQuery lakehouse.
- Fan-out pipeline patterns: single Kafka consumer branching into multiple downstream Iceberg and BigQuery tables.
- CDC (Change Data Capture) solution for near real-time data consistency across lakehouse layers.
- Multi-format Kafka message handling: auto-detection and normalization of multiple message schemas into a unified processing flow.
- Event denormalization strategy: complex nested events split into multiple normalized target tables with CDC support.
- Two-layer schema design (Iceberg source + BigQuery refined) with partition and clustering optimization for analytics workloads.
- Iceberg lakehouse migration: Hadoop Catalog → BigLake Managed Storage (BLMS) REST Catalog, standardized across all pipeline projects.
- Data transformers: Confluent Avro deserialization, multi-format auto-detection, timezone-aware timestamp processing, cross-bundle deduplication.
- Comprehensive unit test suites (pytest) covering transformers, API integration, CDC logic, schema validation, and configuration.
- Code quality enforcement: static typing (mypy), linting (ruff), SonarQube, pre-commit hooks.
- GCP infrastructure provisioning via Terraform: GCS, Artifact Registry, BigLake catalog, Secret Manager, BigQuery datasets.
- Multi-stage GitLab CI/CD: automated testing, Docker image builds (Kaniko), Terraform deployment, Dataflow job submission (STG/PROD).
- Automated BigQuery schema management tooling for deployment (creation, migration, backup/restore).
- Cross-team coordination: Kafka platform team (schema alignment, Schema Registry), infrastructure team (Terraform, IAM, BigLake), analytics team (schema design, query optimization).

PROJECTS

Advanced Info Services Public Company Limited (AIS)

1. Recommendation Engine Project:
 - Profiled and explored data to support the Data Science team in training recommendation models.
 - Designed and developed pipelines to serve data for model prediction.
2. Churn Project:
 - Designed and developed pipelines to serve data for churn model prediction.

Zimple Media Company Limited

1. Propertyhub:
 - Developed the frontend web application for information assets (apartments and condos) using React and Next.js.
2. Blog RentHub:
 - Developed a web blog platform using WordPress.

SCB Data-X

1. Regulatory Data Transformation (RDT):
 - Designed and developed data models using PySpark and PySpark SQL on Azure Databricks.
 - Concluded data validation processes.
 - Developed automated tests and generated scripts using PySpark, Databricks Workspace API, and Data Factory API.
 - Designed and developed schema synchronization processes using PySpark on Databricks to ensure data consistency across databases and data warehouses.
2. Regulatory Data Transformation (RDT CardX):
 - Conducted POCs for the Databricks data platform migration to new versions using Unity Catalog.
 - Developed RBAC processes to enhance data security.
 - Profiled data sources in Kafka and Confluence for integration into new platforms.

PROJECTS

NTT DATA — The 1 Central Group

1. Real-Time Event Streaming Pipelines (Kafka → Iceberg → BigQuery)
 - Streaming data pipelines: Confluent Kafka → Apache Beam on Dataflow → two-layer lakehouse (Iceberg + BigQuery).
 - CDC solution for data consistency in the refined layer. Multi-format Kafka message handling (JSON, Avro/Schema Registry) with auto-detection.
 - Tech: Python, Apache Beam, Dataflow, Kafka (Confluent), Avro/Schema Registry, Iceberg (BigLake BLMS), BigQuery CDC, Terraform, GitLab CI/CD
2. Batch Data Ingestion Pipelines (API / PostgreSQL → Iceberg → BigQuery)
 - Batch pipelines: daily incremental loads from REST APIs and PostgreSQL → Iceberg + BigQuery lakehouse.
 - Table partitioning and clustering strategies optimized for analytics workloads.
 - Tech: Python, Apache Beam, Dataflow, PostgreSQL, Iceberg (BigLake BLMS), BigQuery, Cloud Scheduler, Terraform
3. Customer Profile Streaming Pipeline (Pub/Sub → Bigtable → Multi-Destination)
 - Streaming pipeline: Pub/Sub events enriched with Cloud Bigtable lookups → BigQuery (CDC), AWS S3 (Parquet), and Iceberg.
 - Data reconciliation framework for pipeline migration validation.
 - Tech: Python, Apache Beam, Dataflow (Flex Templates), Pub/Sub, Bigtable, BigQuery CDC, AWS S3, Iceberg (BigLake), Cloud Composer (Airflow)

EDUCATION

Bachelor of Science in Applied Mathematics

King Mongkut's Institute of Technology Ladkrabang

June 2013 - June 2016